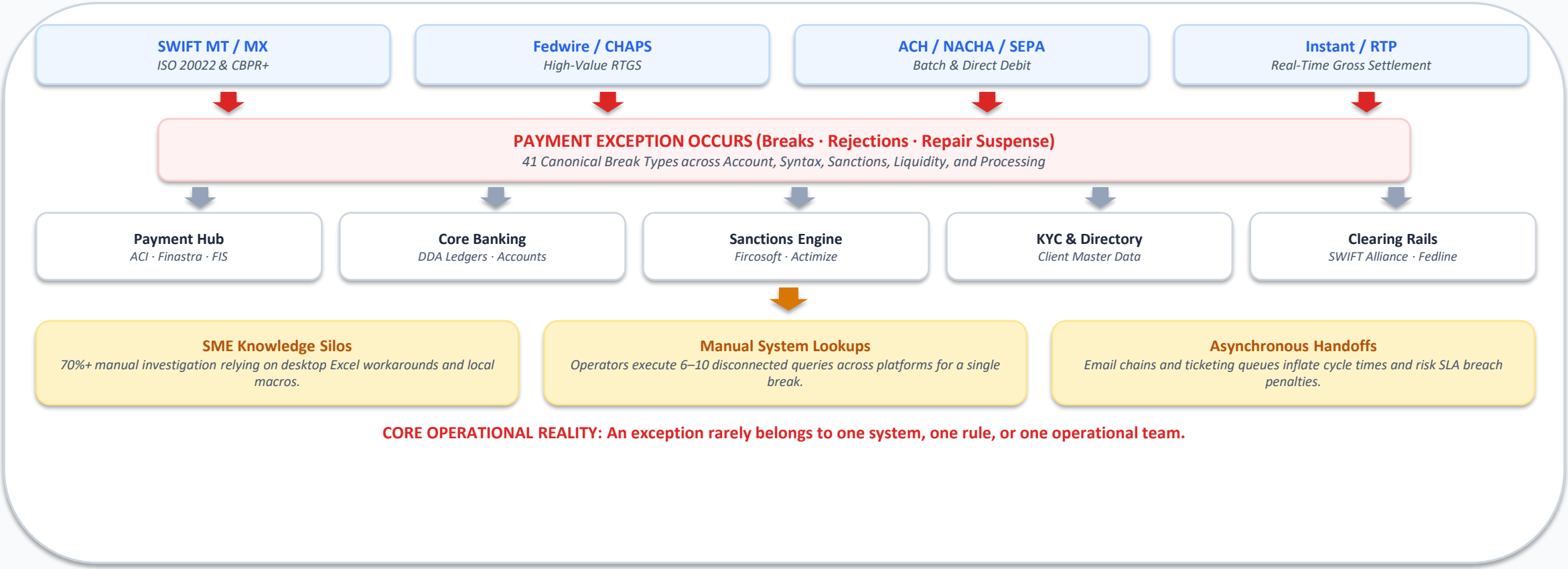
PAGE 1 · THE BUSINESS PROBLEM

Why Banks Need PeX

Payment exceptions are a resolution problem — not simply a transaction-processing problem.



Payment engines process transactions. They do not inherently provide contextual resolution intelligence when execution breaks.

PeX creates a common, governed resolution-intelligence layer across those silos — bridging detection, investigation, and execution.

What Is PeX?

A structured Payment Exception Resolution intelligence layer — built around 41 exception profiles.

REPRESENTATIVE RESOLUTION JOURNEY: INVALID IBAN BREAK (PROFILE #03)

1. Ingest & Detect Break
Raw pacs.008 fails clearing; invalid beneficiary IBAN check digit (MOD-97).

2. Canonical Classification
Classified into Account & Routing Profile #03 (Invalid IBAN/BBAN Syntax).

3. Evidence Corroboration
Asynchronously queries Client Master & SWIFTRef ICDs; retrieves account history.

4. AI Hypothesis Reasoning
Model Gateway synthesizes evidence and proposes valid IBAN with 98% confidence.

5. Deterministic Validation
Rule Engine re-runs ISO 7064 MOD-97-10 check digit math — 100% verified match.

6. Maker-Checker Authorization
Operator reviews side-by-side evidence; supervisor authorizes repair ($0 SoD).

7. Controlled Egress & Settlement
Generates pacs.008 amendment; dispatches via bank rail; pacs.002 ACCP settles case.

WHY PeX IS DIFFERENT: STRUCTURED INTELLIGENCE

41 Canonical Exception Profiles
Comprehensive coverage across Account, Format, Sanctions, Liquidity, & Duplicates.

Live Peripheral Evidence
Queries real bank DDA, KYC, Sanctions, and reference data via secure ICDs.

Deterministic Precedence
Deterministic rules ALWAYS override AI suggestions — zero unverified execution.

Human-in-the-Loop Governance
Hard $0 Maker-Checker segregation of duties; Dual-Auth for transfers ≥$250k.

SHA-256 Merkle Provenance
Every decision node is cryptographically sealed in an immutable audit ledger.

CORE DISTINCTION
"PeX is not an LLM sitting on payment data.
It is a structured resolution capability built around a defined exception universe."

PAGE 3 · HERO LOGICAL ARCHITECTURE

How PeX Resolves a Payment Exception

From exception event to governed resolution and final business outcome.

ZONE 1 · RESOLUTION INTELLIGENCE

1. Ingest & Normalize
Maps raw MT/MX to canonical PaymentExceptionCase

2. 41-Profile Classify
Derives root-cause exception profile & missing fields

3. Evidence Corroboration
Queries DDA, Sanctions, KYC, & SWIFTRef ICDs

4. AI Model Gateway
Generates advisory ReasoningResult JSON & scores

ZONE 2 · DETERMINISTIC CONTROL

5. Deterministic Rules
100% scheme validation — Rules ALWAYS override AI

6. Sanctions Re-Screening
Automated watchlist re-check on proposed repairs

7. Operator Review (HITL)
Side-by-side evidence, rules, & AI rationale review

8. Maker-Checker Approval
Hard $0 SoD limit; Dual-Auth for transfers ≥$250k

ZONE 3 · CONTROLLED EXECUTION

9. Egress Contract Gen
Generates schema-compliant ISO pacs.008 amendment

10. Clearing Dispatch
Dispatches via bank rail; logs transport ACK (DISPATCHED)

11. Final Settlement
Settles ONLY upon pacs.002 ACCP or DDA confirmation

12. Merkle Audit Sealing
Cryptographically seals complete lifecycle into WORM log

CORE ARCHITECTURAL MANTRA: **AI recommends. Controls validate. Humans authorize. Systems execute.**

Closed-Loop Governance: EVENT → INTELLIGENCE → DETERMINISTIC CONTROL → HUMAN AUTHORIZATION → EXECUTION → SETTLEMENT → MERKLE AUDIT

How PeX Fits Around the Bank

Non-intrusive integration. Bank-controlled data. Technology-stack neutrality.

NON-INTRUSIVE INTEGRATION: SURROUNDING THE EXISTING ESTATE

EXISTING BANK ESTATE (Core Banking · Payment Hubs · Clearing)
Finastra, ACI, FIS, Temenos, SAP, Thought Machine, SWIFT Alliance, Fedline

↓

STANDARDIZED ICD & ADAPTER LAYER
REST APIs | Enterprise MQ (IBM/MSMQ) | Event Streams (Kafka/Bus) | Batch SFTP

↓

READ-ONLY EVIDENCE
KYC & Sanctions
Client Master
Reference Data

PeX RESOLUTION CORE
41 Canonical Profiles
Deterministic Rules
HITL Maker-Checker

CONTROLLED EGRESS
Payment Gateways
DDA Ledger Posting
SWIFT / Clearing

↓

BANK-CONTROLLED EXECUTION & SETTLEMENT
PeX never moves funds directly — all repair messages execute via Bank Clearing Gateways.

PeX DOES NOT REQUIRE: Core banking replacement · Payment hub overhaul · Clearing restructuring

PeX ADDS: Common exception semantics · Cross-system evidence · Governed repair · Audit

SECURITY ENCLAVE & ZERO-PII DATA BOUNDARY

Bank Sensitive Data
Raw Names, IBANs, Tax IDs, & Account Numbers

↓

Token Vault Enclave
Dedicated FIPS 140-2 Level 3 KMS CMK Isolation

↓

Surrogate Tokens (TKN_...)
PeX Core processes tokenized records exclusively

↓

Model Gateway Isolation
Advisory LLM prompts receive ZERO raw customer PII

↓

Approved AI Providers
AWS Bedrock · GCP Vertex AI · Azure OpenAI

TECHNOLOGY STACK NEUTRALITY

- Core: Mainframe (z/OS), Distributed On-Prem, Private Cloud, Hybrid
- Rails: SWIFT MT/MX, ISO 20022, Fedwire, ACH, Instant / RTP
- Cloud: AWS, Google Cloud, Azure, On-Prem (100% Feature Parity)
- Data Sovereignty: Bank data NEVER used to train foundation models.

PAGE 5 · ENTERPRISE VALUE & SCALE

Why PeX Can Be Trusted and Scaled

Governed. Traceable. Reusable across heterogeneous bank environments.

1. TRUST · CRYPTOGRAPHIC INTEGRITY

11-Stage Provenance
Every break is tracked in an unbroken cryptographic hash chain.

7-Year WORM Storage
Audit logs stored in unalterable compliance object storage.

Audit Reconstruction
Instant decision reconstruction for internal & regulatory audits.

2. CONTROL · DETERMINISTIC AUTHORITY

Maker-Checker ($0 SoD)
Strict segregation of duties; operator who repairs cannot approve.

Dual-Authorization
High-value transfers (≥$250k) mandate two supervisor signoffs.

Deterministic Override
Banking rules ALWAYS override AI suggestions with 100% authority.

3. REUSE · CANONICAL ONTOLOGY

41 Exception Profiles
Comprehensive taxonomy covering all BIAN break categories.

Canonical Semantics
Normalizes disparate legacy representations (Bank A, B, C).

Reusable Patterns
Resolution logic authored once is instantly shared across desks.

4. SCALE · ENTERPRISE EXPANSION

Multi-Entity Growth
Scale across global subsidiaries, hubs, & correspondent branches.

Sub-Linear Cost
Dramatically expand exception volumes without linear headcount.

Cloud & Stack Neutral
Portable across AWS, GCP, Azure, or On-Premises with zero lock-in.

THE EXECUTIVE PROPOSITION

"PeX turns fragmented payment exceptions into a common, governed resolution capability — using AI where it adds intelligence, deterministic controls where they must remain authoritative, humans where accountability is required, and bank-controlled interfaces for execution."

Common resolution intelligence. Bank-controlled execution. Full traceability.